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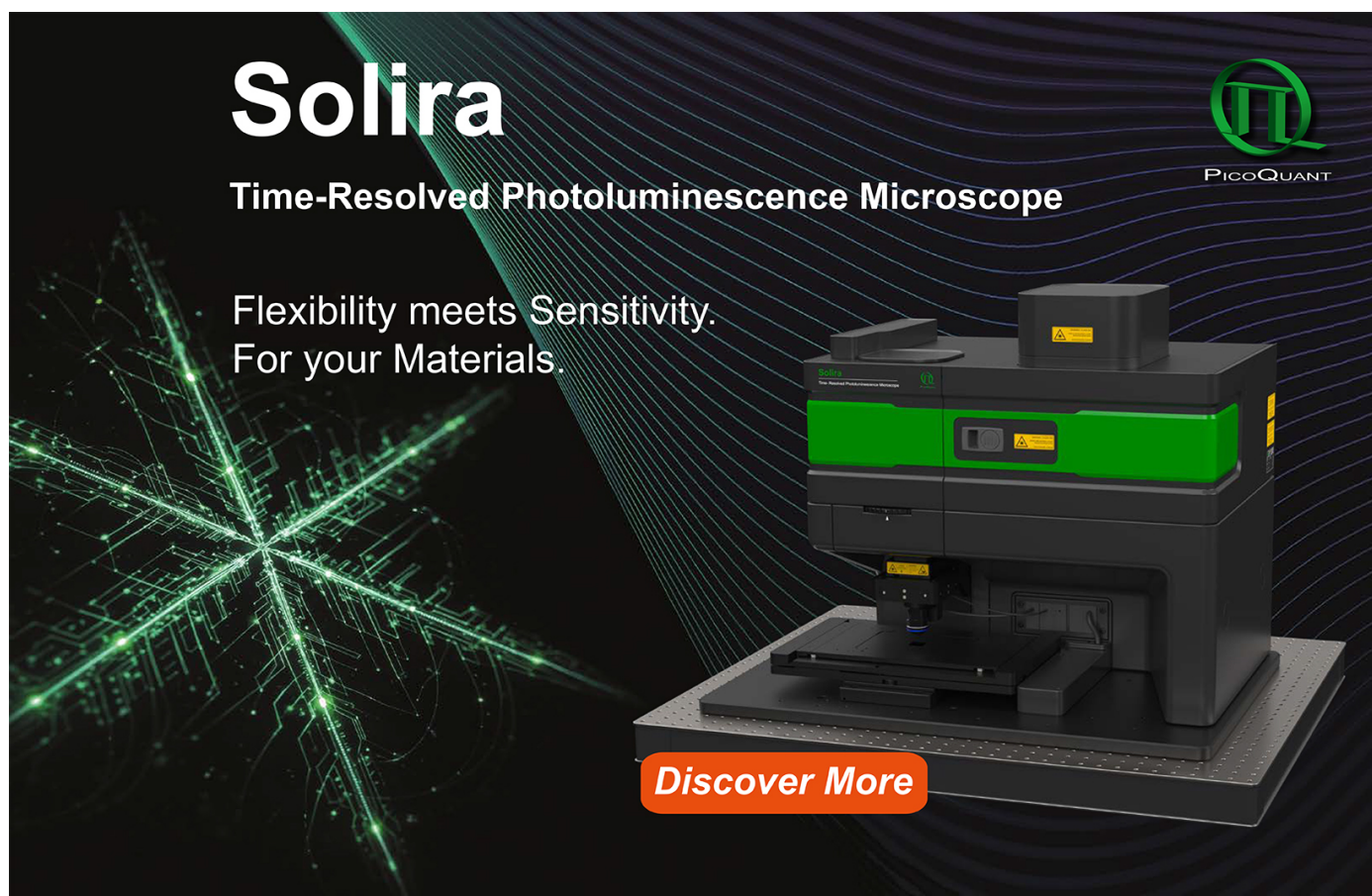
Integration and evaluation of a drone-mounted optical system for river surface velocity measurement

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Integration and evaluation of a drone-mounted optical system for river surface velocity measurement

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Abstract

Large-scale particle image/tracking velocimetry using drones is an evolving method for measuring surface flow in natural environments, providing access to areas challenging for human operators. However, current applications face two main limitations: the challenge of controlling drone positioning over large rivers without ground control points in the field of view (FOV), and insufficient understanding of measurement sensitivity to flight parameters such as hover height and stability. This study proposes an integrated measurement system featuring dual cameras and three lasers. The lasers serve as guiding points in the FOV, where changes in measurement height or orientation alter the laser tracers' positions in images, which makes it possible to estimate the acquired image's position and rotation relative to the water surface. A laboratory experiment assessed the sensitivity and accuracy of the proposed camera-laser configuration in measuring the surface distance and tilt angle. Results show that the system accurately estimates the acquired image's position and orientation relative to the measured surface. Uncertainty analysis shows that system uncertainty depends on both measurement height and laser centroid detection precision, with high precision $\sigma_{\text{pix}} = 0.6$ pixel.

1. Introduction

Measuring flow velocity is essential for discharge estimation. River velocities vary both daily due to hydropower operations and human activities, and seasonally due to natural factors such as snowmelt, rainfall patterns, and storm events. For operational safety and efficiency in hydropower production, water flow velocity helps monitor approach velocities to the intake structure, optimize turbine operations, and identify potential hazards for maintenance work (Bytčanková *et al* 2021, Sinha 2022). Flow discharge measurements are essential to assess power generation potential, develop prediction methods, and support emergency management. Understanding river flow is crucial for environmental assessments, including evaluation of aquatic habitats, fish passage conditions, pollutant transport, river erosion, and debris movement (Govers 1992, Förstner 2004, Nestler *et al* 2008, Piper *et al* 2015). Measuring these flows in the vicinity of hydropower facilities, however, presents challenges due to the vast and complex geometry of these structures. Traditional monitoring methods for surface flows rely on invasive techniques using current meters (Tazioli 2011), and acoustic Doppler instrumentation (Yorke and Oberg 2002, Mueller *et al* 2009). Previous studies have demonstrated the effectiveness of these measurement methods. However, their applications are limited to small streams and easily accessible locations. To overcome these challenges, significant research and development efforts have focused on non-contact measurement techniques, such as radar systems (Costa *et al* 2006) and thermal infrared imaging (Puleo *et al* 2012). While these techniques have significantly advanced hydrological monitoring, they have limitations due to high costs and time-consuming implementation and are often impractical for frequent monitoring of water bodies. To address these limitations, researchers have developed and applied various image-based techniques for flow analysis over the past decades (Fujita *et al* 1998, 2007, Muste *et al* 2008).

Among the optical methods, large-scale particle image velocimetry (LSPIV), particle tracking velocimetry (PTV), and space-time Image velocimetry (STIV) (Fujita *et al* 2007) are the most widely used for remote surface flow measurement systems. The cameras are typically stationary and positioned at a distance from the water surface. The camera is adjusted to point directly downward and is consistently kept perpendicular to the water surface. Surface flow velocity is estimated by applying surface particle tracking algorithms to sequences of images. In scenarios where the camera is inclined relative to the water surface (Hauet *et al* 2008, Eltner *et al* 2020), acquired images must be orthorectified using ground control points (GCPs) within the field of view (FOV). After orthorectification, metric assignment and surface velocity estimation are processed similarly to nadir camera cases. Kim (2006), Hauet *et al* (2008), and Kantoush *et al* (2011) indicate that LSPIV measurements are highly sensitive to surface particle density, the distribution and accuracy of acquired GCPs, and the image orthorectification process. More recently, Bodart *et al* (2024) showed that operator-dependent processing parameters (frame skip factor, output grid node spacing, and velocity filters) and the surface velocity coefficient are additional key sources of variability in LSPIV discharge measurements. Trieu *et al* (2021) and Trieu *et al* (2024) addressed the limitations related to image orthorectification and the lack of calibration through a measurement procedure in which multiple cameras are installed on a bridge, along with a reference frame equipped with ground GCPs to calibrate the camera system. The non-intrusive system provides simultaneous measurements of 3D surface velocity, water level variations, and surface fluctuations. The system can be installed beneath bridges to function as a gauge station. However, the measurement area is spatially limited, and expanding coverage requires relocation and reinstallation of the setup at new locations. The requirement for GCPs within the FOV makes water flow data unavailable for unmonitored locations, such as large rivers and river estuaries.

To address the limitations related to spatial constraints and the need for reinstallation when expanding or changing the measurement location, over the past two decades, unmanned aerial systems (UASs) have emerged as popular and versatile tools for studying surface water and fluvial environments. Drones, or UAS, offer significant benefits as platforms for camera equipment. They can access difficult-to-access areas and capture nadir video footage of extensive study regions. Recent studies have explored the efficiency of using UAS to measure surface water velocity by comparing various software packages based on different velocimetry methods (LSPIV, PTV, and STIV) used to analyze UAS images (Koutalakis *et al* 2019). Tauro *et al* (2015) and Tauro *et al* (2016) developed a custom-built commercial drone for fully remote surface flow observations on small-scale streams. In their study, photometric calibration is achieved remotely through a four-laser system installed at the corners of the fuselage, aligned with the drone's yaw axis. The calibration and image acquisition during measurements are conducted using a single GoPro Hero 3 camera oriented with its axis perpendicular to the water surface. Alternative approaches establish camera position and orientation using GPS (Global Positioning System)/IMU (Inertial Measurement Unit) data, optionally combining with a laser range finder to measure distance to the surface for direct geo-referencing (Legleiter *et al* 2024). Such instrumentation, however, is not necessarily lighter or simpler in practice: Legleiter *et al* (2024) required a heavy-lift hexacopter platform without a stabilizing gimbal and still relied on manual GCPs to refine the initial geo-referencing before image-derived velocity estimates could be compared to field measurements for accuracy assessment. Such approaches typically depend on survey infrastructure, such as RTK ground control or manual selection of tie points from an existing base orthophoto, which is not always available. Furthermore, an IMU reports the attitude of the drone airframe, not the local orientation of the water surface, which can differ due to waves, local slope, or superelevation in bends. This surface-induced tilt cannot be recovered from an IMU regardless of its accuracy, and reliable use requires rigorous calibration on a per-flight basis—a step that Legleiter *et al* (2024) noted had not been performed for their tactical-grade IMU. For cameras subject to platform motion, this process requires an image stabilization step to register consecutive frames before particle tracking, ensuring the resulting displacement reflects water motion rather than platform motion. This is typically achieved using stationary features within the image sequence (Pizarro *et al* 2022).

Surface particle characteristics significantly influence surface velocimetry accuracy (Dal Sasso *et al* 2018, 2020, 2021, Trieu *et al* 2024). The study by Tauro *et al* (2016) demonstrated the feasibility of drone-based flow velocimetry using a single camera with laser pointers for calibration. However, green lasers' detectability diminishes with distance and is affected by floating particles, limiting their application in single-camera systems. While gimbals enable nadir imaging in most drone-based flow observations, stabilizing multi-sensor systems remains challenging due to flight dynamics, GPS accuracy, payload balance, and wind conditions. Accurate image geolocation and alignment are particularly challenging in dynamic environments with limited GCPs or distinct features (Vélez-Nicolás *et al* 2021).

To address the challenges of surface flow observations without GCPs, we develop an integrated camera-laser-based sensing system for autonomous river surface velocity measurements from drones. It combines two synchronized cameras and three lasers on a single platform: one camera with a band-pass filter detects laser reference points, while the second captures flow imagery. The components are mounted on a custom T-shaped frame, ensuring rigid geometric relationships during flight. This paper focuses on evaluating the effectiveness of the proposed system configuration in estimating the distance and tilt angle of the acquired image relative to the measured surface. Through a series of laboratory experiments, we assess how these estimations influence the accuracy of velocity measurements.

Flying heights reported in UAS-based river velocimetry studies vary considerably depending on the sensing technique applied. Optical image velocimetry methods, which rely on resolving discrete surface features, are typically applied at heights of 9–112 m, with accuracy peaking near 62 m and degrading by 112 m due to reduced spatial resolution (Liu *et al* 2021, Wijaya *et al* 2023). UAS-borne Doppler radar, in comparison, has been applied at heights as low as 1.5–6.1 m (Zhou *et al* 2024). Reach-scale mapping approaches that do not rely on resolving discrete surface features, such as thermal image mapping referenced through GPS/IMU and laser range finder data, can instead be applied at heights up to 124–270 m above ground level (Legleiter *et al* 2024). The height range evaluated in this study (3.5–9.5 m) was selected to suit the laboratory test setup and the camera-laser system configuration, as discussed further in section 4, along with strategies to extend the approach to higher operational heights through future work.

The remainder of this paper is structured as follows: section 2 introduces the camera-laser measurement configuration, with the system overview detailed in section 2.1, followed by the theoretical measurement principles in section 2.2 and calibration procedures in section 2.3. Section 2.4 presents the laser centroid estimation method used in the experimental evaluation. Section 3 presents the evaluation of the camera-laser sensing configuration, while section 4 discusses the results and section 5 presents the concluding remarks.

2. Camera-laser measurement configuration, measurement principles, and calibration

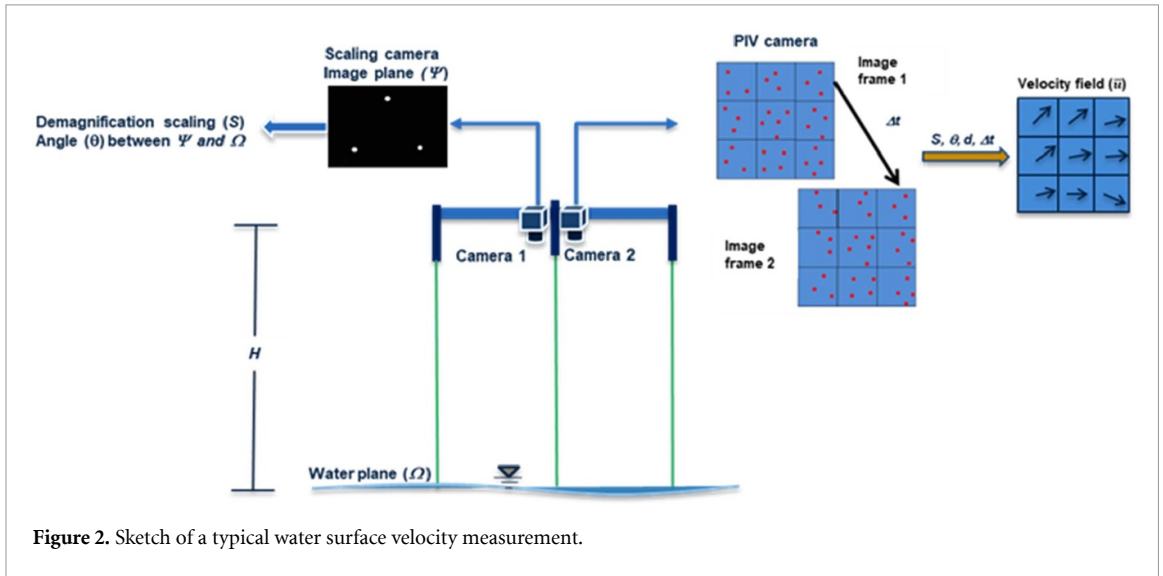
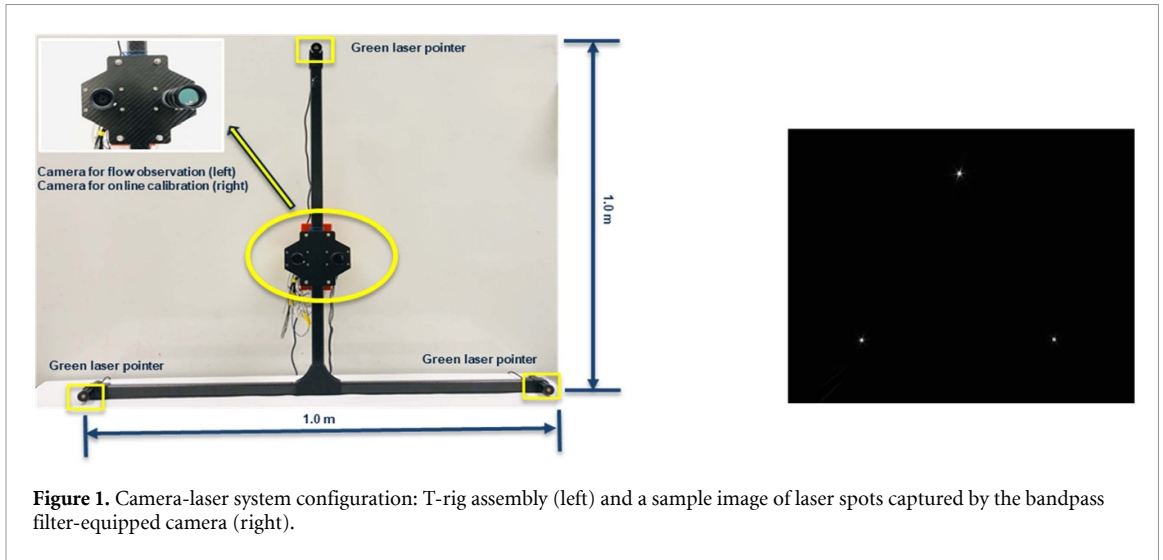
2.1. Camera-laser measurement system overview

The sensing system consists of two cameras and three green laser pointers (wavelength of 532 nm and power of 60.0 mW). To limit potential exposure, the lasers are oriented so that they project onto the water surface directly below the platform rather than outward, and are switched on only during image acquisition, rather than continuously during flight. During testing, personnel near the laser beams wore safety glasses rated for the 532 nm wavelength, and operations were carried out under supervision. The cameras, Genie-Nano-5 G, have a resolution of 2464×2056 pixels and a 16 mm objective lens. The system enables synchronous image capture at various frame rates, with a maximum acquisition speed of 80 frames per second. Two cameras are employed to ensure consistent detection of laser tracers. One camera, equipped with a 532 nm narrow-band interference filter, isolates laser spots from background features, while the other captures the flow surface.

A carbon fiber T-shaped rig serves as the mounting structure for the measurement system components. The setup orients both cameras and lasers downward toward the water surface. Figure 1 illustrates the configuration of cameras and lasers (left), and an example image of laser spots captured by the bandpass filter-equipped camera (right). Three non-collinear laser points are the minimum required to reconstruct the observed plane orientation in 3D, and are mounted at the endpoints of the T-rig, forming a triangle. The two cameras are positioned near its centroid. This geometric configuration ensures that laser tracers remain within the cameras' FOV for flying heights above 3.0 m while maintaining equal radial distances from the image center. This arrangement provides uniform radial distortion, consistent perspective effects, balanced geometric strength, and equal illumination effects under uniform lighting conditions.

2.2. Theory and measurement principles

A sketch of a typical measurement situation is shown in figure 2. Particles flow in a plane, Ω , at a velocity $\bar{u}$ [m s^{-1}] and are monitored by the PIV camera (camera 2), having its corresponding image plane Ψ oriented at an angle, θ , relative to Ω . The PIV camera that is positioned at a distance, H , above the water surface captures snapshot images with a time separation Δt that are analyzed to give the displacement field, $\bar{d}$, between two recordings with a random error δd . This gives an apparent velocity $\bar{v} = \bar{d}/\Delta t$ [pixels s^{-1}] in the plane of the detector. To translate the apparent velocity into true velocity, we need to



scale according to the equation,

$$\bar{u} = \frac{S \cdot \bar{v}}{\cos \theta}, \quad (1)$$

where S (m pixel⁻¹) is the demagnification scaling between detector coordinates and world coordinates in image plane Ψ . Equation (1) has two unknowns, S and, which are provided by the scaling camera, mounted next to the PIV camera in the same plane. In addition, we also sought to estimate the precision of this approach and evaluate the sensitivity to different potential sources of error. We differentiate equation (1) to get,

$$\delta |\bar{u}| = \frac{\partial |\bar{u}|}{\partial S} \delta S + \frac{\partial |\bar{u}|}{\partial \theta} \delta \theta + \frac{\partial |\bar{u}|}{\partial d} \delta d = |\bar{u}| \left[\frac{\delta S}{S} + \tan \theta \delta \theta + \frac{\delta d}{|\bar{d}|} \right], \quad (2)$$

where the magnitudes of the velocities are introduced for convenience. Assume the random errors in S , θ , and $\bar{d}$ are mutually uncorrelated, with standard deviations σ_s , σ_θ and σ_d . Then independent components combine in quadrature, equivalent to a Euclidean norm. We may therefore express the expected uncertainty in velocity $|\bar{u}|$ as,

$$\sigma_{|\bar{u}|} = \frac{S \cdot |\bar{v}|}{\cos \theta} \sqrt{\left(\frac{\sigma_s}{S}\right)^2 + (\tan \theta \sigma_\theta)^2 + \left(\frac{\sigma_d}{|\bar{d}|}\right)^2} = |\bar{u}| \sqrt{\left(\frac{\sigma_s}{S}\right)^2 + (\tan \theta \sigma_\theta)^2 + \left(\frac{\sigma_d}{|\bar{d}|}\right)^2}. \quad (3)$$

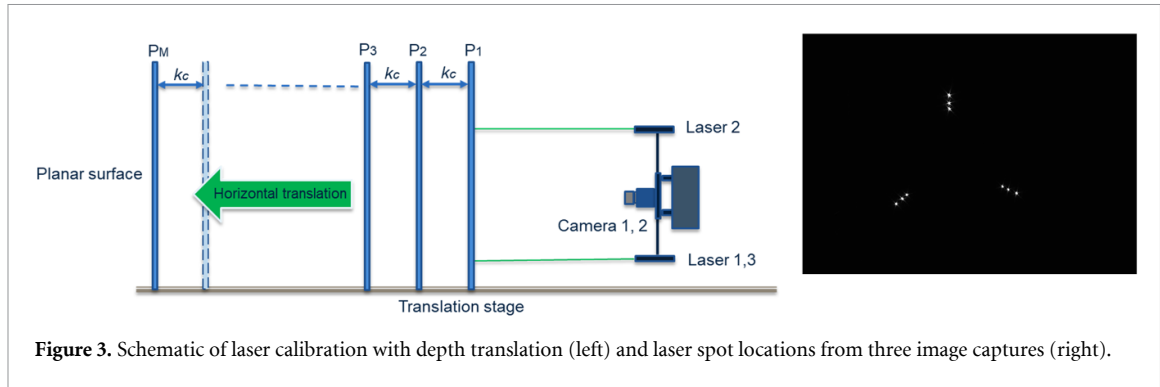


Figure 3. Schematic of laser calibration with depth translation (left) and laser spot locations from three image captures (right).

The uncertainty propagation described by equations (1)–(3) proceeds through the following sequence: uncertainty in laser centroid detection (section 2.4) first propagates, through the camera-laser triangulation, into uncertainty in the estimated 3D coordinates of the laser spots on the water surface; these coordinate uncertainties then propagate into uncertainty in the estimated water plane, that is, the demagnification, S and tilt angle θ describing the surface's distance and orientation relative to the camera-laser system; and finally, the uncertainties in S and θ propagate into the velocity uncertainty through equation (3).

Comparing with equation (1), we see that the combined standard deviation $\sigma_{|\vec{u}|}$ of the true velocity $|\vec{u}|$ scales with $|\vec{u}|$ itself and relative uncertainties of the demagnification, angle, and apparent particle motion, respectively. While the uncertainty in the motion estimation, δd , depends on specifics of the PIV algorithm used, the uncertainties, δS , in demagnification, and $\delta\theta$, in angle, respectively, depend on with what precision the centroid of each laser spot can be detected in the scaling camera and on the reliability of the camera calibration. In the following subsections, we will first introduce the process used to calibrate the two cameras to minimize the influence of systematic errors and then analyze random errors caused by uncertainties in the centroid estimation. These findings are then evaluated in section 3 through a series of controlled experiments.

2.3. Calibration procedures

The geometric calibration process involves two cameras and three lasers, all mounted in fixed relative positions to each other, as described in section 2.1. First, the two cameras are calibrated to determine the intrinsic and extrinsic camera parameters. Subsequently, the three lasers are calibrated to establish their positions and directions within the same coordinate system as the cameras.

The camera calibration process is performed using a checkerboard calibration plate. This pattern was positioned at 44 different arbitrary locations and orientations within the measurement volume, at distances ranging from 3.5 m to 6.5 m, with the initial position and orientation defining the global coordinate system for the cameras. The intrinsic and extrinsic camera parameters were determined from the detected calibration pattern, following the methods described in Trieu *et al* (2021).

Following camera calibration, the next step is determining the equations that describe the positions and direction vectors of the three laser pointers within the defined coordinate system. The calibration process is sketched in figure 3. A flat white painted plate is placed perpendicularly in front of the camera system and is moved in steps of k_c throughout the measurement volume. At each position, an image of the three laser dots is acquired by the scaling camera, providing information of the apparent motion of the laser spots.

Each laser point in 3D space can be represented as the intersection of the laser ray and the camera ray, which extends from the measurement volume through the lens to the camera sensor. This 3D point must lie on a plane parallel to the initial planar surface at P_1 , separated by a distance k_i . The total translation distance at capture P_i (where $i = 1, 2, \dots, M$) is denoted as $k_i = (i - 1)k_c$, with $k_1 = 0$ representing the initial position at P_1 . In practice, the translation may deviate from perfect perpendicularity to the planar surface. Therefore, to achieve accurate calibration, two-directional parameters $n_k(n_1, n_2)$ of the translation must be estimated. Each laser beam requires 4 parameters to define its position and direction in space. With three lasers in the measurement system, the total number of laser parameters for calibration becomes 12. Thus, the complete calibration requires the estimation of 14 parameters, where the two additional parameters account for the directional components. The parameter estimation problem



Figure 4. Example of the field test setup for examining laser tracer visibility.

can be formulated as:

$$\min_{\{n_k\}\{E_j\}} \sum_{i=1}^M \sum_{j=1}^N \Gamma(\chi^{(i,j)}, \Upsilon^{(i,j)}, k_i, n_k, E_j) \quad (4)$$

where $E_j \in \mathbb{R}^4$ represents the vector containing position and direction parameters for the laser j , and Γ is a function of square residuals. These residuals are computed between two 3D points: χ , the intersection point of the camera ray and laser ray, and Υ , the corresponding point on the translated planar surface. The residuals are calculated at each translation step.

The laser parameters are estimated using the objective function (4). A trust-region method is employed to iteratively improve parameter values from their initial values. This process is implemented using MATLAB's *fminunc* function from the optimization toolbox. Explicit expressions of the gradients and Hessian for the objective function in (4) are provided to define the trust region. To assess the accuracy of the camera-laser calibration, reprojection error is used, which represents the geometric deviation between detected laser spots in the images and their corresponding projected points using estimated camera-laser parameters. This depth translation calibration method achieves a mean reprojection error of 0.1 pixels.

2.4. Centroid estimation

The visibility and shape of laser tracers on the water surface are influenced by the laser-to-water distance and water transparency. Accurate detection of the laser tracer's center location is crucial for achieving precise measurements. Therefore, we examined the laser tracer characteristics under different conditions. Figure 4 shows an example of the field test setup to examine laser tracer visibility. Here, we investigated laser tracer visibility and shape at three measurement distances (3.5 m, 6.0 m, and 9.5 m) and under three water clarity conditions (muddy, relatively muddy, and clear water).

The laser tracers appear as comet-shaped patterns, with their tails oriented inward toward the center of the triangle formed by the three laser points (see figure 5(a)). Typical examples of spot shapes for various conditions are shown in figure 6. Due to this asymmetrical shape, the detection algorithm comprises two steps: (1) detecting elliptical regions fitted to each laser tracer through thresholding, binarization, and ellipse fitting (see figure 5(b)), and (2) localizing the tracer's center point by searching along the ellipse's major axis, parallel to its minor axis, identifying the location with maximum total intensity, and estimating the center point location using intensity weighting methods (see figure 5(c)). To mitigate peak locking issues, a second-order polynomial is fitted to the peak curve using the nine nearest points with the highest intensity values within the peak region (see figure 5(d)). This sub-pixel refining process is analogous to Gaussian peak-fitting methods used in particle image velocimetry systems, such as PIVlab (Thielicke and Stamhuis 2021).

Given image sequences captured within 0.5 s, the laser tracer locations could be assumed to remain consistent across all frames. To examine this spatial consistency, we detected the center points of each laser tracer throughout the image sequences. Figure 6 (left) shows laser spot detection results from images acquired at three heights (3.5 m, 6.5 m, and 9.2 m) under three water transparency conditions:

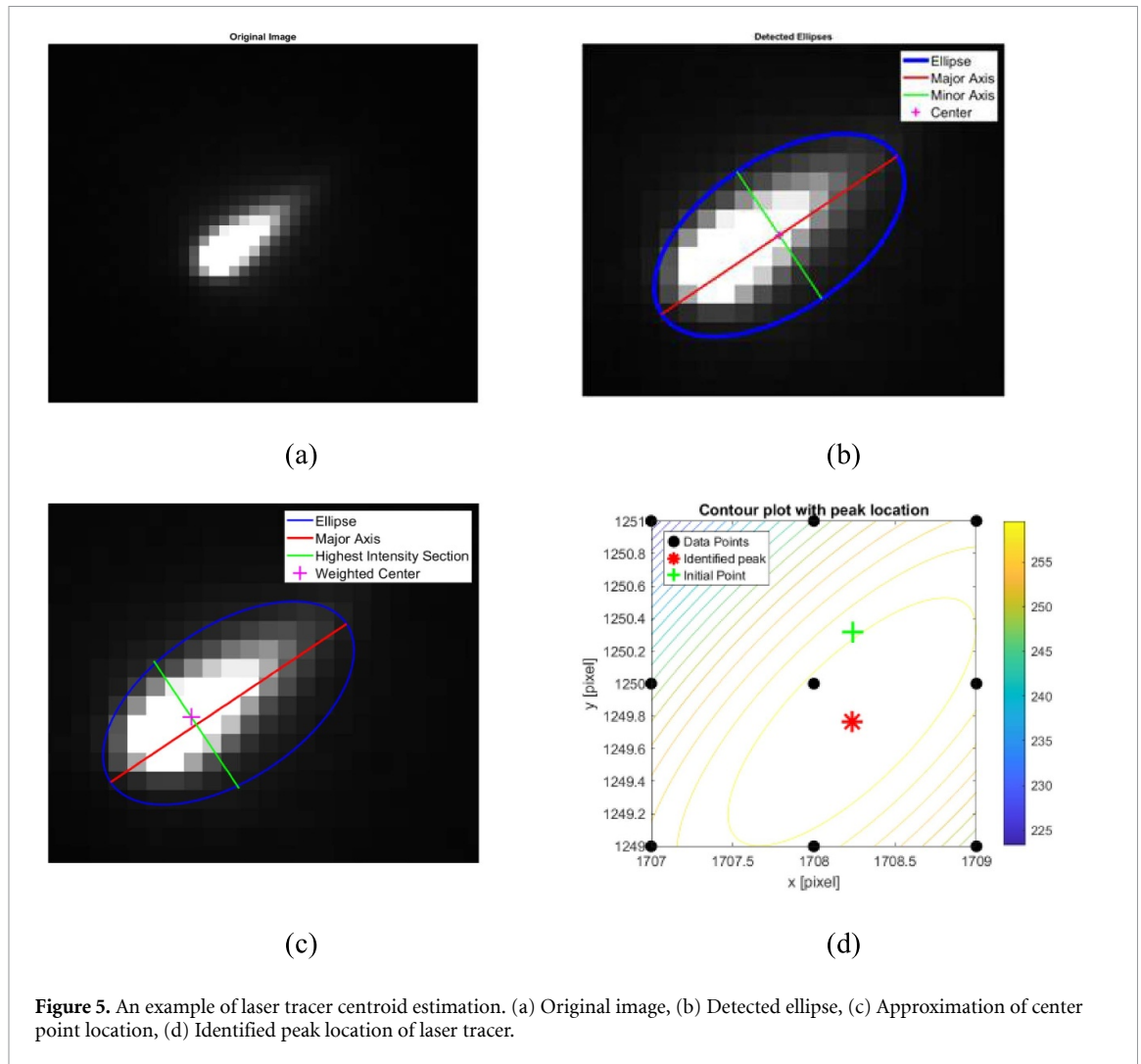


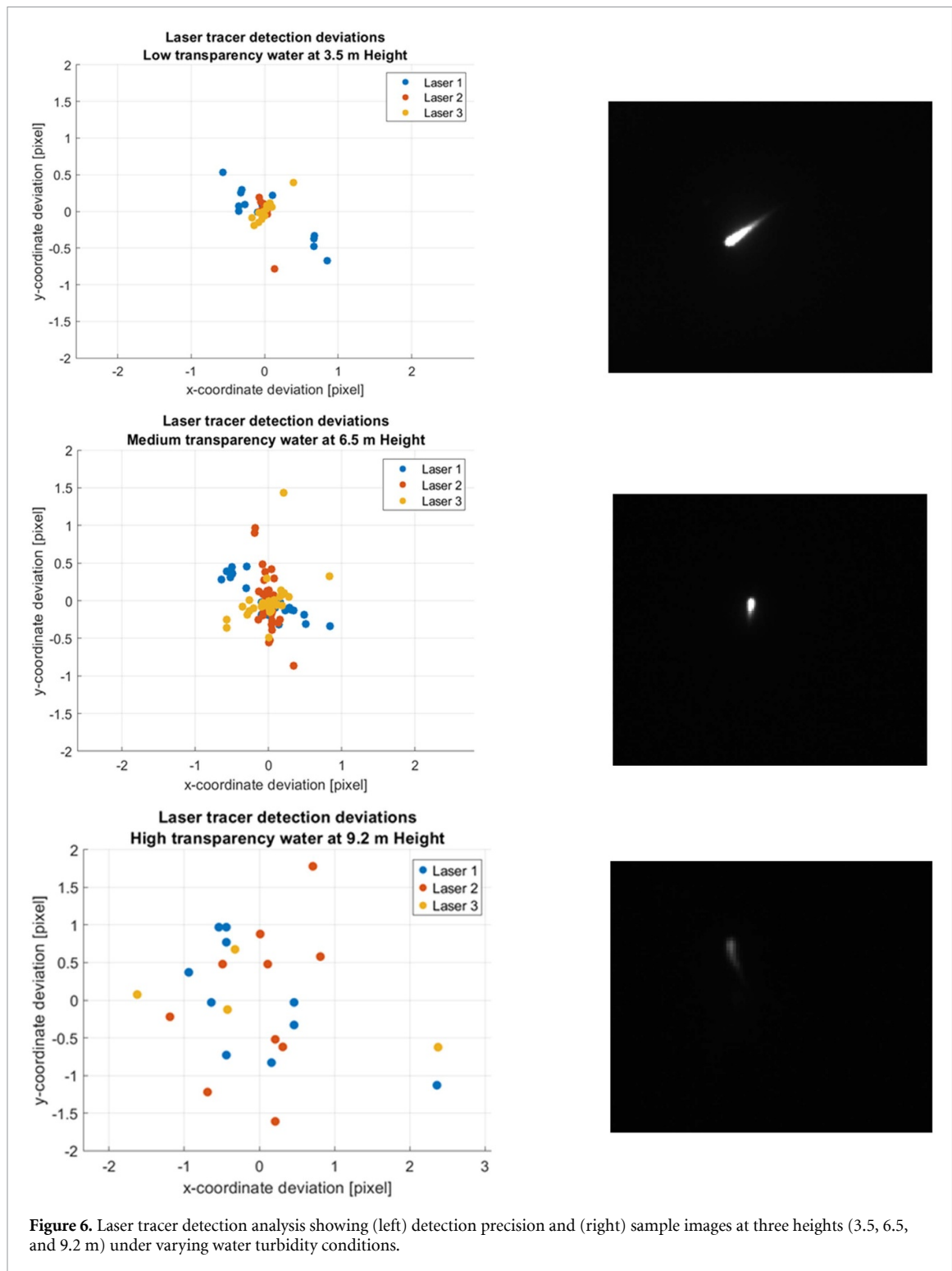
Figure 5. An example of laser tracer centroid estimation. (a) Original image, (b) Detected ellipse, (c) Approximation of center point location, (d) Identified peak location of laser tracer.

low (muddy), medium (moderately muddy), and high (clear water), respectively. The results illustrate the deviation of detected coordinates from their mean positions across the image sequence. The laser tracer detection standard deviations were found to be 0.6, 0.9, and 1.8 pixels for low, medium, and high transparency conditions, respectively.

The field tests described above were all conducted under bright, direct sunlight, including conditions with specular glint/reflection visible on the water surface, representing a challenging outdoor lighting scenario for optical laser detection. As shown in the sample images in figure 6 (right), the bandpass-filter-equipped camera effectively suppressed both the broadband ambient sunlight and the specular glint, such that only the laser tracers remained visible in the acquired images. This test confirms that the narrow-band filtering approach reliably isolates the laser spots from the background even under bright sunlight and glint conditions, supporting the robustness of the laser detection method for outdoor field deployment. The three field tests did not vary independently in terms of distance, water transparency, ambient lighting, or wind and surface-wave conditions. Each distance had a different turbidity condition, and the 9.2 m test had high wind and surface ripples. The stated detection standard deviations (0.6, 0.9, and 1.8 pixels) are a result of multiple factors and should be interpreted accordingly.

3. Evaluation of the camera-laser sensing configuration

Laboratory evaluation was conducted to assess the sensitivity and optimize the performance of the proposed camera-laser configuration. The evaluation focused on analyzing system performance at various heights and tilt angles relative to the measured surface, as flight height is a critical parameter for velocity measurements. In most aerial platform surface flow measurements, the camera is typically mounted on a



gimbal to maintain the nadir orientation of the optical axis relative to the water surface. This configuration ensures constant metric dimensions across all pixels in the captured images. In this study, the evaluation methodology combines theoretical principles with reference measurements to provide a reliable assessment of the proposed camera-laser system configuration. The magnification analysis specifically focuses on images acquired at nadir.

3.1. Laboratory experiments and results

The camera-laser configuration described in section 2 is evaluated through laboratory experiments, where camera-laser-based estimations of demagnification and tilt angle are validated against theoretical predictions and experimental reference values. The setup is sketched in figure 7. Due to vertical

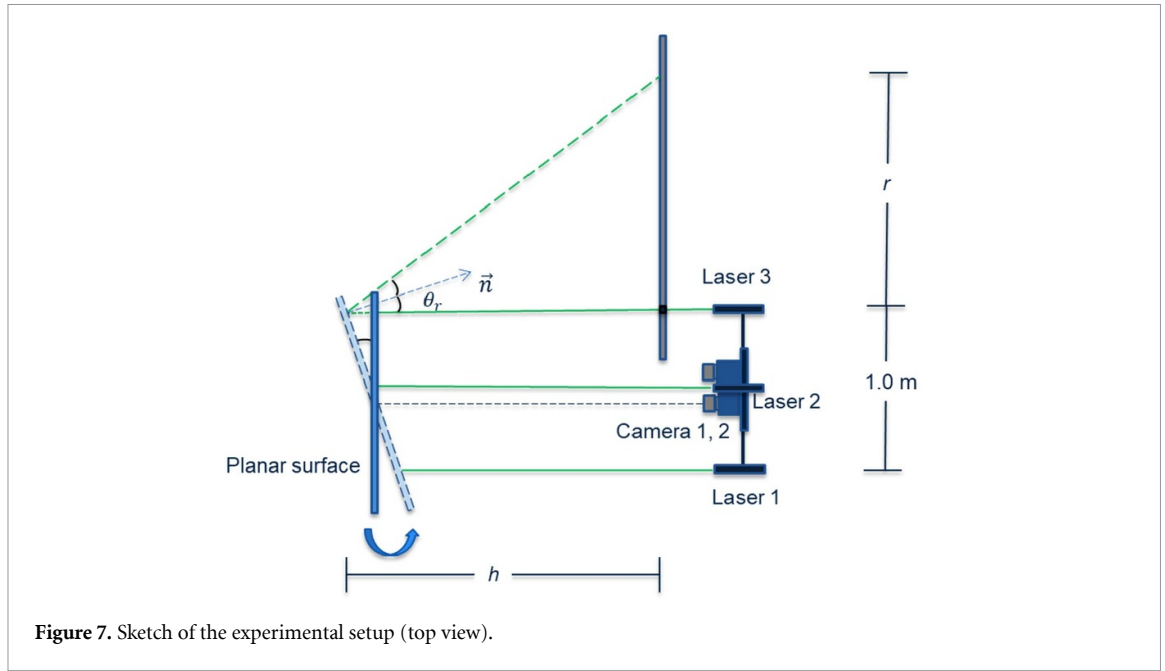


Figure 7. Sketch of the experimental setup (top view).

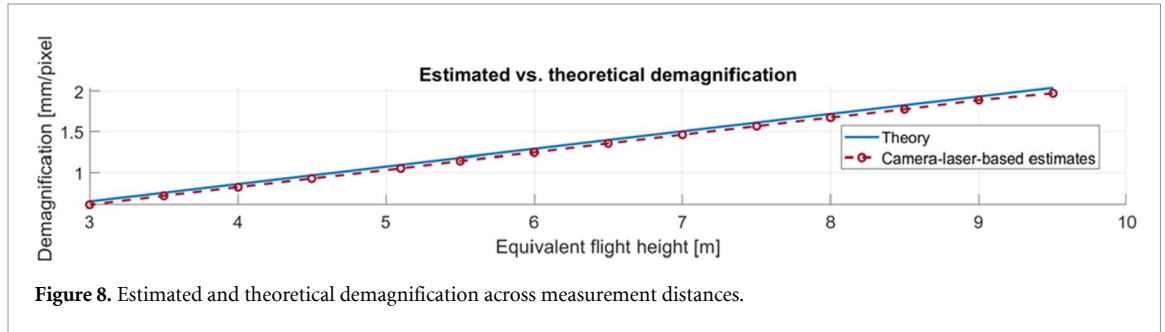


Figure 8. Estimated and theoretical demagnification across measurement distances.

space constraints, the setup was arranged horizontally to simulate equivalent conditions of varying flight heights. The theoretical demagnification is calculated as,

$$s = \frac{L}{\Delta} \quad (5)$$

where L (mm) is physical distance measured between laser spots on the plate and Δ (pixel) is the corresponding distances in the images. A laser arrangement, including a hole, was used to adjust the tilt angle θ_r at the plate. The tilt angle, θ_r (rad) is determined using the measured displacement r (m) and the distance h (m) between the hole and the planar surface, according to the equation:

$$\theta_r = \frac{1}{2} \arctan \left(\frac{r}{h} \right). \quad (6)$$

The angle measured through this experimental setup serves as the reference value for the validation of the tilt angle θ_c estimated from the scaling system.

The observed planar surface is translated horizontally from 3.5 to 9.5 m to assess the sensitivity of magnification at different distances. At specific distances, tilt angles are introduced to the planar surface to evaluate the system's accuracy in estimating surface orientations.

Figure 8 presents the results from the translation test, where the camera-laser system was moved along the distance range of 3.5 – 9.5 m. The red dashed line represents the camera-laser-based demagnification estimates (S), while the blue solid line shows theoretical demagnification for nadir-acquired images.

Figure 9 compares the camera-laser-based estimated tilt angles, θ_c , with the experimental reference values, θ_r , calculated from equation (6), with different colors representing each equivalent flight height. The red dashed line indicates perfect agreement between methods.

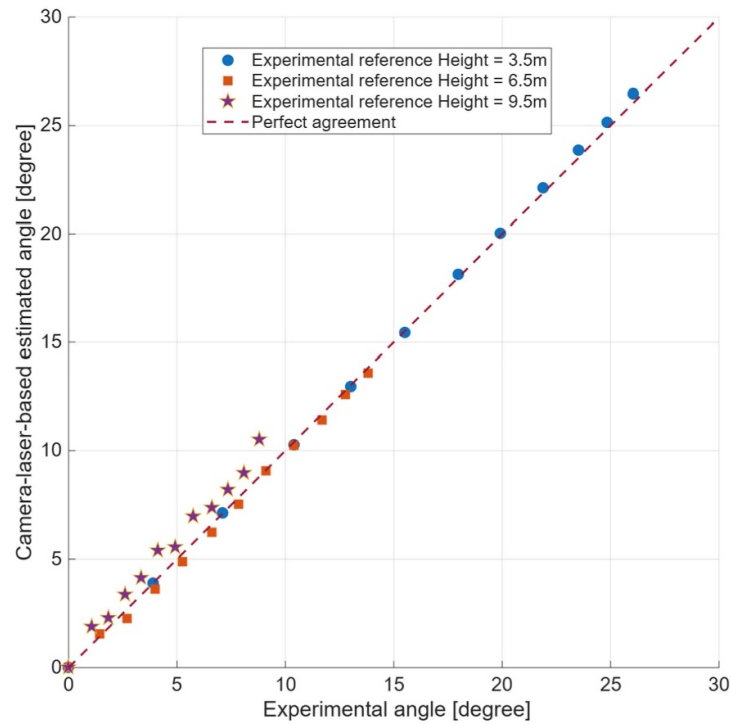


Figure 9. Comparison of camera-laser-based angle measurements and experimental references at 3.5 m, 6.5 m, and 9.5 m equivalent flight heights.

Table 1. Mean reprojection error of reconstructed 3D laser positions, lateral error, depth-direction error, and tilt estimation error at three equivalent heights, inside (3.5 m, 6.5 m) and outside (9.5 m) the calibrated range.

Equivalent height (m)	Mean projection error (pixel)	Mean error		
		Lateral error (mm)	Depth direction error (mm)	Tilt estimation error (degree)
3.5	0.668	2.21	13.40	0.15
6.5	0.933	2.82	17.90	0.24
9.5	1.132	5.76	40.55	0.86

To further quantify the tilt angle estimation accuracy inside and outside the calibrated range, mean reprojection error of the reconstructed 3D laser positions, lateral error, depth-direction error, and tilt estimation error were computed at the three equivalent heights (3.5 m, 6.5 m, and 9.5 m) shown in figure 9, and are summarized in table 1.

3.2. Relative errors in velocity estimation

In this section, simulations are used to estimate the relative contributions of uncertainties in demagnification and tilt angle to the uncertainty in the velocity estimate. Equation (3) contains three parameters: demagnification, S , tilt angle, θ , and motion estimation, d . The relative uncertainty contribution (RU_i) of each parameter was determined by

$$RU_i(\%) = \frac{e_i}{\sigma_{|u|}/|\bar{u}|} \cdot 100 \quad (7)$$

where $e_i = \frac{\sigma_s}{s}$, $\tan\theta\sigma_\theta$, $\frac{\sigma_d}{|d|}$, respectively. For this analysis, $\frac{\sigma_d}{|d|}$ was assumed fixed, as this parameter is independent of laser centroid detection precision. Camera-laser calibration determines each laser beam's position and orientation in 3D space. At each measurement height, the laser intersects the water surface, reprojected as a point in the camera image. As height varies, these image points align along a straight line in the image plane. Therefore, detection uncertainties of laser points are constrained to lie along this reprojected line. For the simulation, Gaussian random errors with standard deviations of 0.6, 0.9, and 1.8 pixels (corresponding to low, medium, and high transparency conditions from section 2.4) were

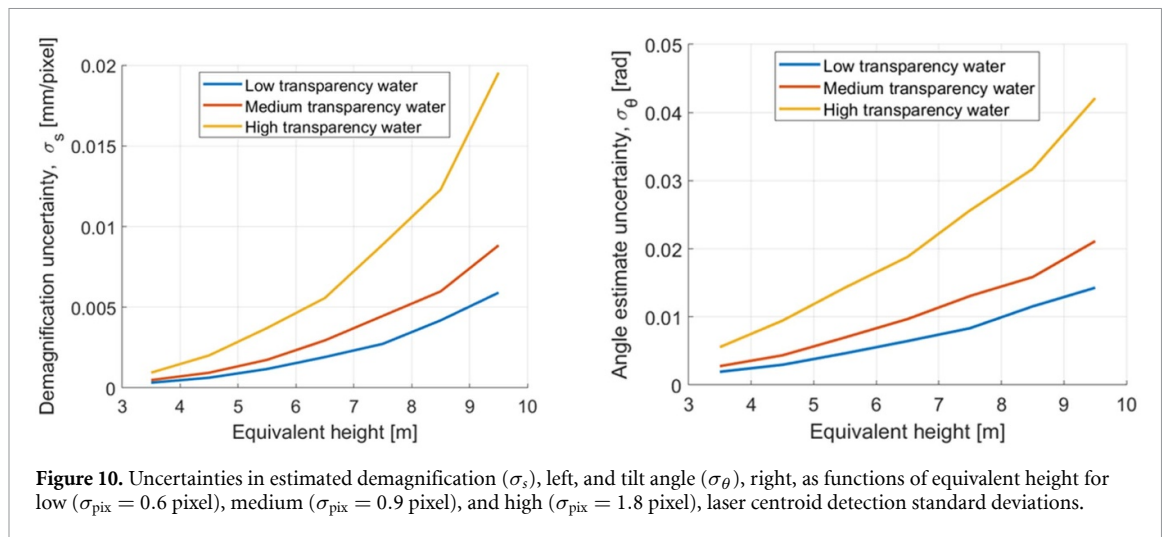


Figure 10. Uncertainties in estimated demagnification (σ_s), left, and tilt angle (σ_θ), right, as functions of equivalent height for low ($\sigma_{\text{pix}} = 0.6$ pixel), medium ($\sigma_{\text{pix}} = 0.9$ pixel), and high ($\sigma_{\text{pix}} = 1.8$ pixel), laser centroid detection standard deviations.

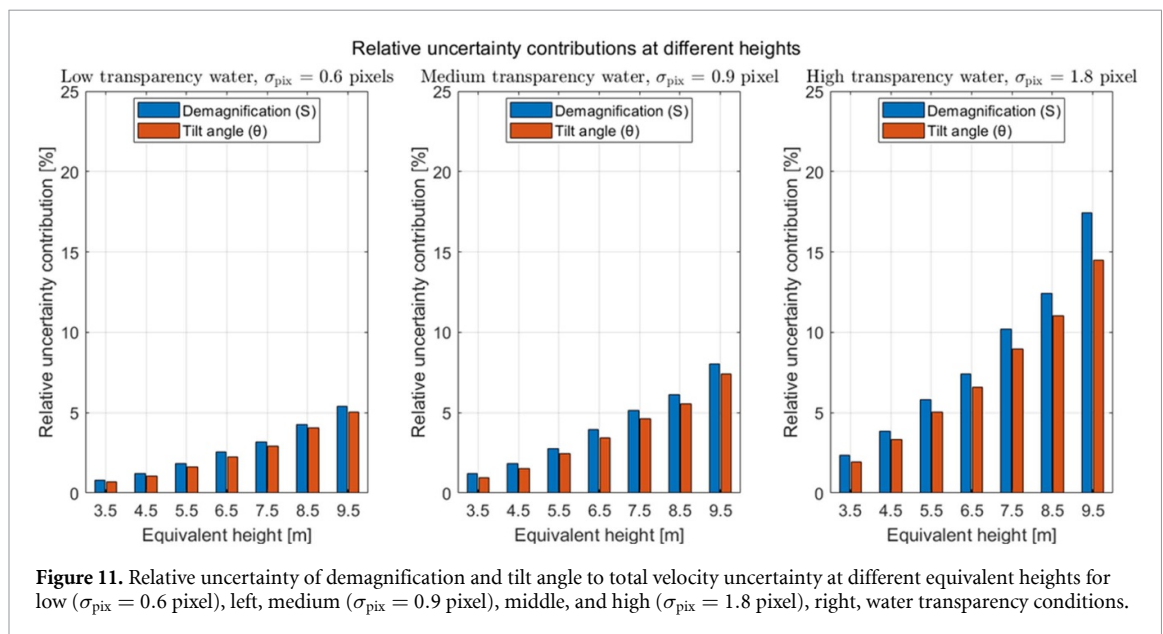


Figure 11. Relative uncertainty of demagnification and tilt angle to total velocity uncertainty at different equivalent heights for low ($\sigma_{\text{pix}} = 0.6$ pixel), left, medium ($\sigma_{\text{pix}} = 0.9$ pixel), middle, and high ($\sigma_{\text{pix}} = 1.8$ pixel), right, water transparency conditions.

added to the true laser centroid coordinates along their respective laser lines. These perturbed coordinates were processed through the calibrated camera-laser system to reconstruct 3D positions via triangulation. The resulting 3D coordinates were used to compute the demagnification factor and the plane defined by the three laser points. To focus on the system-dependent uncertainties under various height and detection precision conditions, the motion estimation parameters were fixed at, $|d| = 1.0$ pixel, and $\sigma_d = 0.05$ pixel (5.0% relative displacement uncertainty), representing a worst-case scenario with small particle displacement. The uncertainties in estimated demagnification and tilt angle were determined by comparing perturbed estimates with true values. Simulations were performed for all three detection standard deviations (0.6, 0.9, and 1.8 pixels) at heights from 3.5 to 9.5 m. Figure 10 presents the resulting uncertainties in demagnification and tilt angle as functions of height. Figure 11 shows only the relative contributions of S and θ to emphasize system-dependent uncertainties.

4. Discussion

The demagnification estimated by the camera-laser configuration demonstrates good agreement with the theoretical demagnification shown in figure 8. The values of demagnification increase linearly as the equivalent flight height increases. Figure 9 demonstrates a strong correlation between experimental and camera-laser-based measurements, showing that the camera-laser configuration accurately estimates the planar surface tilt angle. The system shows higher accuracy at lower heights (3.5 m and 6.5 m).

At 9.5 m equivalent height, the estimated angles show a consistent trend of slightly overestimating tilt angles, suggesting the presence of a systematic error. This deviation seems to occur because the 9.5 m height exceeds the system's calibrated range of 3.5–6.5 m. Table 1 quantifies this effect. Within the calibrated range (3.5 m, 6.5 m), all error metrics increase moderately with height. At 9.5 m, beyond the calibrated range, error growth accelerates markedly, particularly for depth direction error and tilt estimation error. Depth-direction error is 6–7 times larger than lateral error at all three heights because triangulation inherently estimates distance to the surface less precisely than it estimates lateral position; this gap widens as height increases. Since tilt angle is derived from depth differences between the three laser points, tilt angle error follows the same accelerating trend as depth-direction error, explaining the tilt angle overestimation observed at 9.5 m. In field applications, this out-of-range degradation can be avoided by acquiring calibration images at the intended operational flying height, using the drone-mounted system, rather than relying on a fixed laboratory-calibrated range. Performing calibration at, or close to, the height at which flow measurements are to be conducted would ensure the calibrated volume covers the actual measurement height.

Figure 6 shows that the laser tracer visibility and detection precision are height dependent. The system achieves distinct laser tracer visibility at a low altitude (3.5 m) with tightly clustered detection points ($\sigma_{\text{pix}} = 0.6$ pixel). At the medium altitude (6.5 m), while laser tracers remain visible, detection points show moderately increased scatter ($\sigma_{\text{pix}} = 0.9$ pixel). At 9.2 m, reduced tracer visibility leads to significant scatter in detection points (up to $\sigma_{\text{pix}} = 1.8$ pixel). Notably, measurements at 9.2 m were conducted under high wind conditions, where high surface ripples likely affected laser incidence angles. These increased incidence angles may reduce tracer visibility due to enhanced light scattering and reflection at the water surface. Figure 10 shows the demagnification uncertainty (left) and tilt angle uncertainty (right) as functions of equivalent height under three water transparency conditions. Both uncertainties increase nonlinearly with height across all conditions, with the rate of increase becoming more pronounced at higher heights, particularly for high-transparency water. High laser detection uncertainty ($\sigma_{\text{pix}} = 1.8$ pixel) produces the largest errors in estimated demagnification and tilt angle. This occurs because the laser beam penetrates below the water surface and scatters, which blurs the laser spot boundary and reduces the precision of centroid detection. The resulting centroid error propagates through the camera ray model into the estimated 3D coordinates of the laser spots, and subsequently into the demagnification and tilt angle of the water plane estimated from those three points.

Figure 11 shows the relative contributions of demagnification (S) and tilt angle (θ) uncertainties to total velocity uncertainty at different measurement heights and laser detection precision levels. Both contributions increase with height across all transparency conditions, with demagnification consistently contributing slightly more than tilt angle. At lower detection precision ($\sigma_{\text{pix}} = 1.8$ pixel), the relative contributions reach approximately $RU_S = 18\%$ and $RU_\theta = 15\%$ at 9.5 m, which are roughly 3–4 times larger than those at higher detection precision ($\sigma_{\text{pix}} = 0.6$ pixel) at the same height. This demonstrates that the precision of laser centroid detection has a critical influence on overall system uncertainty, particularly at greater heights, where a small centroid detection error translates into a proportionally larger error in the estimated 3D laser point positions, amplifying uncertainties in both demagnification and tilt angle. Notably, the relative uncertainty contributions at 9.5 m with $\sigma_{\text{pix}} = 0.6$ pixel are approximately equal to those at 7.5 m with $\sigma_{\text{pix}} = 0.9$ pixel, and at 5.5 m with $\sigma_{\text{pix}} = 1.8$ pixel, indicating that improved detection precision can effectively compensate for increased measurement height, maintaining consistent system accuracy across different heights.

At high detection precision ($\sigma_{\text{pix}} = 0.6$ pixel), system-dependent uncertainties remain below 11% of total variance across the entire tested height range, indicating stable and reliable performance. At low detection precision ($\sigma_{\text{pix}} = 1.8$ pixel), however, uncertainty grows rapidly above 7.0 m, reaching 33% of total random error at maximum height. These results suggest that operational height limits should be decided based on acceptable uncertainty thresholds for specific applications. For instance, applications requiring system uncertainties below 15% of total random error should restrict low precision operations to heights below approximately 7.0 m, whereas high detection precision imposes no significant operational restrictions within the tested range. These values are relative errors; therefore, they will change with higher velocities or lower image acquisition frequencies.

The present evaluation further highlights additional considerations for field implementation. Accurate laser centroid detection is the key factor in achieving reliable performance across the full range of operating heights, and the achievable detection precision should be explicitly considered when defining operational boundaries. In addition, the precision of surface particle motion estimation may remain a significant technical challenge because it depends on both the availability of surface particles (Dal Sasso *et al* 2020, Trieu *et al* 2024) and the stability of the drone platform.

Table 2. Qualitative comparison of the proposed camera-laser configuration with selected existing approaches for river surface velocimetry.

Aspect	GCP-based LSPIV/photogrammetry	UAS-borne Doppler radar	GPS/IMU + laser rangefinder	Camera-laser triangulation(this study)
Positioning/infrastructure	Requires GCPs within the FOV for orthorectification (Lewis and Rhoads 2018, Trieu <i>et al</i> 2021, Trieu <i>et al</i> 2024). Image-based stabilization also needs stationary terrestrial features (Pearce <i>et al</i> 2020, Pizarro <i>et al</i> 2022)	Independent of optical GCPs; radar payload mounted on the UAS (Fulton <i>et al</i> 2020, Zhou <i>et al</i> 2024)	Requires tactical-grade GNSS/IMU, RTK ground control (Essel <i>et al</i> 2022, Legleiter <i>et al</i> 2024). IMU requires per-flight calibration (Legleiter <i>et al</i> 2024, Kinzel <i>et al</i> 2024)	Self-contained; recovers distance and tilt from laser spots on the water surface; no external infrastructure required
Typical operational height	Mid-range optical heights (e.g. 9–112 m); accuracy peaks near 20–62 m and degrades at greater heights (Lewis and Rhoads 2018, Liu <i>et al</i> 2021, Wijaya <i>et al</i> 2023)	Validated at low heights (e.g. 1.5–6.1 m above the water) (Zhou <i>et al</i> 2024, Fulton <i>et al</i> 2020)	Can be applied at high altitude (e.g. 124–270 m AGL) (Kinzel <i>et al</i> 2024, Legleiter <i>et al</i> 2024)	Evaluated at 3.5–9.5 m in laboratory tests; extendable with longer focal length lens
Accuracy considerations	Sensitive to GCP distribution/accuracy, orthorectification quality, and operator-dependent processing parameters (Kim 2006, Hauet <i>et al</i> 2008, Bodart <i>et al</i> 2024, Eltner <i>et al</i> 2020)	Provides surface velocity within radar footprint; less dependent on tracers; affected by propwash at low flight heights and low flow velocities (Zhou <i>et al</i> 2024, Fulton <i>et al</i> 2020)	Depends on GNSS/IMU quality, DEM accuracy, and geo-referencing workflow; automated geo-referencing can produce large misalignment vs manual GCP-based referencing (Legleiter <i>et al</i> 2024, Kinzel <i>et al</i> 2024, Eltner <i>et al</i> 2021)	Depends on laser centroid detection precision and calibration range; validated for distance and tilt angle in the laboratory
Cost/complexity	Camera-based; cost depends on platform and field surveying of GCPs (Lewis and Rhoads 2018, Pearce <i>et al</i> 2020)	Additional radar instrumentation; requires signal processing for propwash separation (Zhou <i>et al</i> 2024, Fulton <i>et al</i> 2020, Kinzel and Legleiter 2019)	Higher instrumentation and survey cost: heavy-lift platform, RTK equipment, and laser rangefinder (Legleiter <i>et al</i> 2024, Kinzel and Legleiter 2019)	Dual-camera and three-laser payload on a custom T-rig; no external survey infrastructure required
Robustness	Limited when GCPs or stationary terrestrial features are unavailable, e.g. wide rivers or full-water FOV (Pizarro <i>et al</i> 2022; Trieu <i>et al</i> 2021; Lewis and Rhoads 2018)	Less affected by optical lighting and tracer visibility; works in turbulent, featureless water (Zhou <i>et al</i> 2024; Fulton <i>et al</i> 2020)	Less dependent on optical surface features; still needs survey infrastructure; IMU cannot recover wave-induced surface tilt (Legleiter <i>et al</i> 2024; Essel <i>et al</i> 2022)	Works when the entire FOV is water; laser detection affected by water transparency, wind, and flight height
Key limitations	Unavailable for wide rivers/estuaries without accessible GCPs; operator-dependent processing parameters add variability (Bodart <i>et al</i> 2024)	Point/line measurement rather than full-field imaging; propwash contaminates signal at low flow velocities (Zhou <i>et al</i> 2024, Fulton <i>et al</i> 2020)	IMU cannot recover local water-surface tilt; per-flight IMU calibration required but often not performed; infrastructure not always available (Legleiter <i>et al</i> 2024, Kinzel <i>et al</i> 2024, Essel <i>et al</i> 2022)	Requires visible laser spots on the water surface; performance degrades outside the calibrated height range; field validation ongoing

Table 2 provides a qualitative comparison of the proposed camera-laser configuration with GCP-based LSPIV/photogrammetry (e.g. Lewis and Rhoads 2018, Eltner *et al* 2020, Pearce *et al* 2020, Liu *et al* 2021, Trieu *et al* 2021, 2024, Wijaya *et al* 2023), UAS-borne Doppler radar (Kinzel and Legleiter 2019,

Fulton *et al* 2020, Zhou *et al* 2024), and GPS/IMU with laser rangefinder (Eltner *et al* 2021, Essel *et al* 2022, Kinzel *et al* 2024, Legleiter *et al* 2024), in terms of positioning requirements, typical operational height, accuracy considerations, cost and complexity, robustness, and key limitations. Since the present study evaluates the camera-laser measurement principle under laboratory conditions, the comparison is qualitative. A quantitative field comparison against these established techniques is part of our ongoing work.

The height range evaluated in this study (3.5–9.5 m) was selected based on the camera FOV combined with the camera-laser system configuration (section 2.1), and represents a range suited to the laboratory test setup rather than an operational limit of the underlying measurement principle. Extending the system to greater flying heights for field applications could be achieved using a longer focal length camera lens, allowing the laser tracers to remain within the FOV at greater distances without changing the underlying triangulation approach. This would preserve the core advantage of the three-laser triangulation configuration, namely the simultaneous estimation of surface distance and tilt angle from a single image, while extending the system's applicable height range. The proposed integrated camera-laser system could be readily applied for autonomous river surface velocity measurements from drones. For rivers wider than the camera's FOV, the system can be operated at multiple stationary positions spanning the channel width, with the drone held in place during image acquisition at each position before moving to the next. The resulting sub-area velocity fields can then be merged into a single cross-sectional measurement, consistent with multi-position acquisition strategies used in related image-based river velocimetry studies. Applying and validating this strategy for larger rivers is part of our ongoing field research.

5. Conclusions

This study presented and evaluated a camera-laser configuration that estimates both surface distance and tilt angle from a single image, without requiring external position sensors or GCPs. Across the tested height range (3.5–9.5 m), the system reliably estimates image distance and orientation relative to the measured surface. Demagnification uncertainty consistently contributes slightly more to total measurement uncertainty than tilt angle uncertainty across all tested conditions. Both measurement height and laser centroid detection precision significantly influence system uncertainty, with their effects being strongly interdependent. High detection precision ($\sigma_{\text{pix}} = 0.6$ pixel) maintains system-dependent uncertainties below 11% of total random error across the full tested height range, whereas low detection precision ($\sigma_{\text{pix}} = 1.8$ pixel) leads to rapid uncertainty growth above 7.0 m. These findings demonstrate that laser centroid detection precision is the critical parameter governing system reliability, and that operational height limits should be defined accordingly for specific application requirements.

Extending the system to broader field scenarios, including larger rivers and greater flying heights, together with calibration at the intended operational height, is ongoing work aimed at optimizing the performance of drone-based surface velocimetry in the field. This will also include comprehensive testing against established measurement techniques to assess measurement accuracy, system reliability, and operational robustness under real-world conditions.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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Conflict of interest

The authors declare no conflict of interest.

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References

- Bodart G, Le Coz J, Jodeau M and Hauet A 2024 Quantifying and Reducing the Operator Effect in LSPIV Discharge Measurements *Water Resour. Res.* **60** e2023WR034740
- Bytčanková L, Rumann J and Dušička P 2021 Evaluation of the flow velocity distribution in the intake structure of a small hydropower plant *IOP Conf. Ser.: Mater. Sci. Eng.* **1203** 022102
- Costa J, Cheng R, Haeni F, Melcher N, Spicer K, Hayes E, Plant W, Hayes K, Teague C and Barrick D 2006 Use of radars to monitor stream discharge by noncontact methods *Water Resour. Res.* **42** W07422
- Dal Sasso S F, Pizarro A and Manfreda S 2020 Metrics for the quantification of seeding characteristics to enhance image velocimetry performance in rivers *Remote Sens.* **12** 1789
- Dal Sasso S F, Pizarro A, Pearce S, Maddock I and Manfreda S 2021 Increasing LSPIV performances by exploiting the seeding distribution index at different spatial scales *J. Hydrol.* **598** 126438
- Dal Sasso S, Pizarro A, Samela C, Mita L and Manfreda S 2018 Exploring the optimal experimental setup for surface flow velocity measurements using PTV *Environ. Monit. Assess.* **190** 1–14
- Eltner A, Geber M, Guenther F and Schwalbe E 2021 Using thermal and RGB UAV imagery to measure surface flow velocities of rivers *ISPRS Arch.* **43** 717–24
- Eltner A, Sardemann H and Grundmann J 2020 Flow velocity and discharge measurement in rivers using terrestrial and unmanned-aerial-vehicle imagery *Hydrol. Earth Syst. Sci.* **24** 1429–45
- Essel B, McDonald J, Bolger M and Cahalane C 2022 Initial study assessing the suitability of drones with low-cost GNSS and IMU for mapping over featureless terrain using direct georeferencing *ISPRS Arch.* **43** 37–44
- Förstner U 2004 Sediment dynamics and pollutant mobility in rivers: an interdisciplinary approach *Lakes Reserv.: Res. Manag.* **9** 25–40
- Fujita I, Muste M and Kruger A 1998 Large-scale particle image velocimetry for flow analysis in hydraulic engineering applications *J. Hydraul. Res.* **36** 397–414
- Fujita I, Watanabe H and Tsubaki R 2007 Development of a non-intrusive and efficient flow monitoring technique: the space-time image velocimetry (STIV) *Int. J. River Basin Manag.* **5** 105–14
- Fulton J W *et al* 2020 QCam: sUAS-Based Doppler Radar for Measuring River Discharge *Remote Sens.* **12** 3317

- Govers G 1992 Relationship between discharge, velocity and flow area for rills eroding loose, non-layered materials *Earth Surf. Process. Landf.* **17** 515–28
- Hauet A, Creutin J-D and Belleudy P 2008 Sensitivity study of large-scale particle image velocimetry measurement of river discharge using numerical simulation *J. Hydrol.* **349** 178–90
- Kantoush S A, Schleiss A J, Sumi T and Murasaki M 2011 LSPIV implementation for environmental flow in various laboratory and field cases *J. Hydro-Environ. Res.* **5** 263–76
- Kim Y 2006 *Uncertainty Analysis for Non-intrusive Measurement of River Discharge Using Image Velocimetry* (The University of Iowa)
- Kinzel P J and Legleiter C J 2019 sUAS-based remote sensing of river discharge using thermal particle image velocimetry and bathymetric lidar *Remote Sens.* **11** 2317
- Kinzel P J, Legleiter C J, Gazoorian C, Nelson J M, Volber J, Wrathal J and Anderson I 2024 Reach-scale mapping of surface flow velocities from thermal images acquired by an uncrewed aircraft system along the sacramento River, California, USA *Water* **16** 1870
- Koutalakis P, Tzoraki O and Zaimes G 2019 UAVs for hydrologic scopes: application of a low-cost UAV to estimate surface water velocity by using three different image-based methods *Drones* **3** 14
- Legleiter C J, Kinzel P J, Dille M, Vespignani M, Wong U, Anderson I, Hyde E, Gazoorian C and Cramer J M 2024 Mapping river flow from thermal images in approximately real time: proof of concept on the sacramento river, California, USA *Remote Sens.* **16** 4746
- Lewis Q W and Rhoads B L 2018 LSPIV measurements of two-dimensional flow structure in streams using small unmanned aerial systems: 1. accuracy assessment based on comparison with stationary camera platforms and in-stream velocity measurements *Water Resour. Res.* **54** 8000–18
- Liu W-C, Lu C-H and Huang W-C 2021 Large-scale particle image velocimetry to measure streamflow from videos recorded from unmanned aerial vehicle and fixed imaging system *Remote Sens.* **13** 2661
- Mueller D S, Wagner C R, Rehmel M S, Oberg K A and Rainville F 2009 Tazioli, US Department of the Interior, US Geological Survey Reston, (EUA).
- Muste M, Fujita I and Hauet A 2008 Large-scale particle image velocimetry for measurements in riverine environments *Water Resour. Res.* **44** W00D19
- Nestler J, Goodwin R, Smith D, Anderson J and Li S 2008 Optimum fish passage and guidance designs are based in the hydrogeomorphology of natural rivers *River Res. Appl.* **24** 148–68
- Pearce S et al 2020 An evaluation of image velocimetry techniques under low flow conditions and high seeding densities using unmanned aerial systems *Remote Sens.* **12** 232
- Piper A T, Manes C, Siniscalchi F, Marion A, Wright R M and Kemp P S 2015 Response of seaward-migrating European eel (*Anguilla anguilla*) to manipulated flow fields *Proc. R. Soc. B* **282** 20151098
- Pizarro A, Dal Sasso S F and Manfreda S 2022 VISION: vIdeo StabilisatION using automatic features selection for image velocimetry analysis in rivers *SoftwareX* **19** 101173
- Puleo J A, McKenna T E, Holland K T and Calantoni J 2012 Quantifying riverine surface currents from time sequences of thermal infrared imagery *Water Resour. Res.* **48** W01527
- Sinha S 2022 Power intake size optimization for economical hydropower *Results Eng.* **15** 100572
- Tauro F, Pagano C, Phamduy P, Grimaldi S and Porfiri M 2015 Large-scale particle image velocimetry from an unmanned aerial vehicle *IEEE/ASME Trans. Mechatronics* **20** 3269–75
- Tauro F, Porfiri M and Grimaldi S 2016 Surface flow measurements from drones *J. Hydrol.* **540** 240–5
- Tazioli A 2011 Experimental methods for river discharge measurements: comparison among tracers and current meter *Hydrol. Sci. J.* **56** 1314–24
- Thielicke W and Stamhuis E J 2021 PIVlab—Time-Resolved Digital Particle Image Velocimetry Tool for MATLAB *Figshare Dataset* (<https://doi.org/10.6084/m9.figshare.1092508.v19>)
- Trieu H, Bergström P, Sjö Dahl M, Hellström J G I, Andreasson P and Lycksam H 2021 Photogrammetry for Free Surface Flow Velocity Measurement: from Laboratory to Field Measurements *Water* **13** 1675
- Trieu H, Bergström P, Sjö Dahl M, Hellström J G I, Andreasson P and Lycksam H 2024 Natural surface floaters in image-based river surface velocimetry: insights from a case study *Flow Meas. Instrum.* **96** 102557
- Vélez-Nicolás M, García-lópez S, Barbero L, Ruiz-ortiz V and Sánchez-bellón Á 2021 Applications of unmanned aerial systems (UASs) in hydrology: a review *Remote Sens.* **13** 1359
- Wijaya F, Liu W-C, Suharyanto S and Huang W-C 2023 Comparative assessment of different image velocimetry techniques for measuring river velocities using unmanned aerial vehicle imagery *Water* **15** 3941
- Yorke T H and Oberg K A 2002 Measuring river velocity and discharge with acoustic Doppler profilers *Flow Meas. Instrum.* **13** 191–5
- Zhou Z et al 2024 Measuring river surface velocity using UAS-borne Doppler radar *Water Resour. Res.* **60** e2024WR037375